

Report on Cooperative Optimization Project

March 2026

1 Part I

1.1 Decentralized gradient descent and gradient tracking

1.1.1 Justification of the step size

Let denote the agent $A_1, A_2, \dots, A_5$ The function f_i for each agent is:

$$f_i(\alpha) = \frac{\sigma^2}{5} \frac{1}{2} \alpha^T K_{mm} \alpha + \frac{1}{2} \sum_{i \in A_i} (y_i - K_{(i)m} \alpha)^2 + \frac{\nu}{10} \|\alpha\|^2 \quad (1)$$

One has:

$$\nabla_{\alpha} f_i(\alpha) = \frac{\sigma^2}{5} K_{mm} \alpha - \sum_{i \in A_i} K_{(i)m}^T (y_i - K_{(i)m} \alpha) + \frac{\nu}{5} \alpha \quad (2)$$

Then:

$$\nabla_{\alpha}^2 f_i(\alpha) = \frac{\sigma^2}{5} K_{mm} + \sum_{i \in A_i} K_{(i)m}^T K_{(i)m} + \frac{\nu}{5} I_m \quad (3)$$

So, let $\lambda_{i,\max} = \arg \max_{\lambda \in \text{Sp}(\nabla_{\alpha}^2 f_i(\alpha))} |\lambda|$ for each agent i . Let $L = \max_i |\lambda_{i,\max}|$. Then, we need to choose a step size which fulfilled is $\leq O\left(\frac{1}{L}\right)$.

1.1.2 Plot convergence depending on graph structure

Concerning the DGD, we observe the smaller is the error ball, and the better is the consensus. Indeed, one has theoticaly, that the error ball for DGD is of order $O\left(\frac{\alpha}{1-\gamma}\right)$ where γ is the spectral gap of the graph, and α is the step size. We also have an error on the consensus, which is of order $O\left(\frac{\alpha}{1-\gamma}\right)$. The gradient tracking algorithm it reaches zero error ball and zero consensus error.

1.1.3 Visualization of the obtained function

test

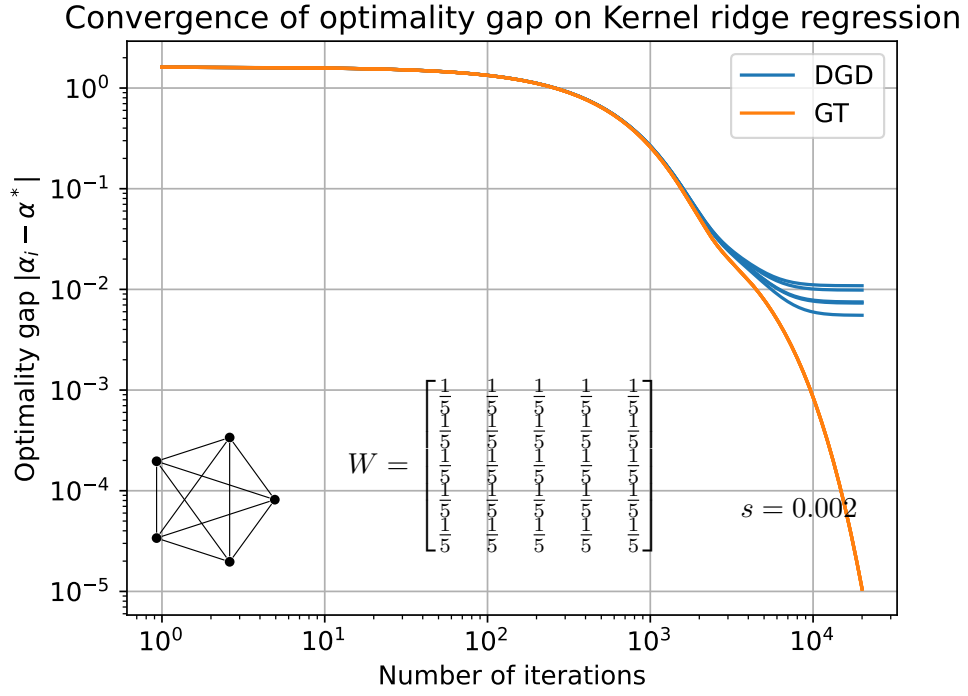


Figure 1: Convergence in terms of $\|\alpha_i - \alpha^*\|$ for DGD and Gradient tracking on the Kernel ridge regression example for a fully connected graph.

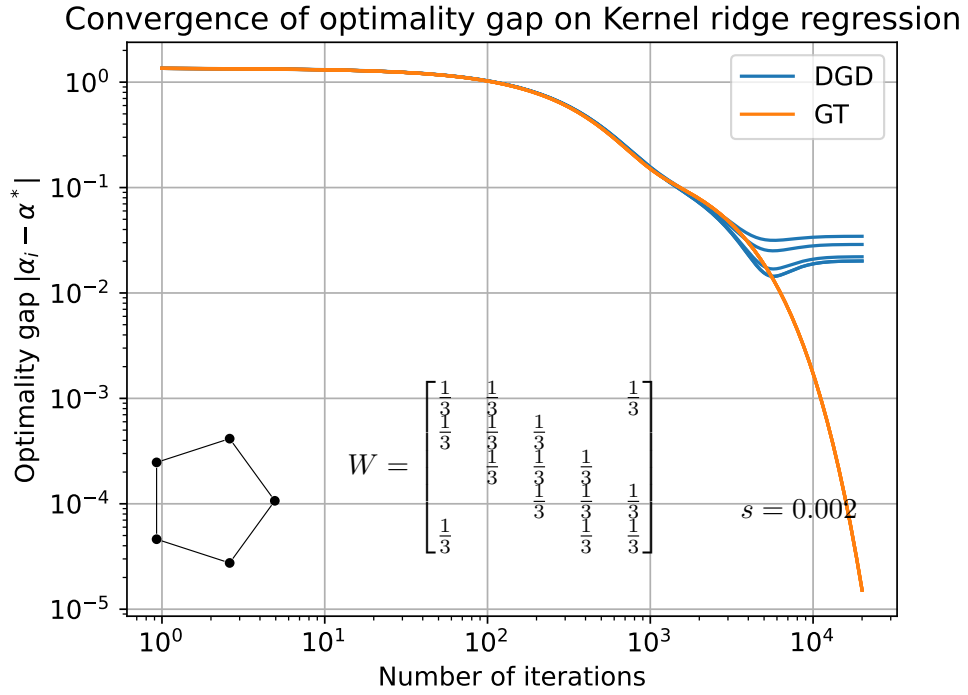


Figure 2: Convergence in terms of $\|\alpha_i - \alpha^*\|$ for DGD and Gradient tracking on the Kernel ridge regression example for a small world graph.

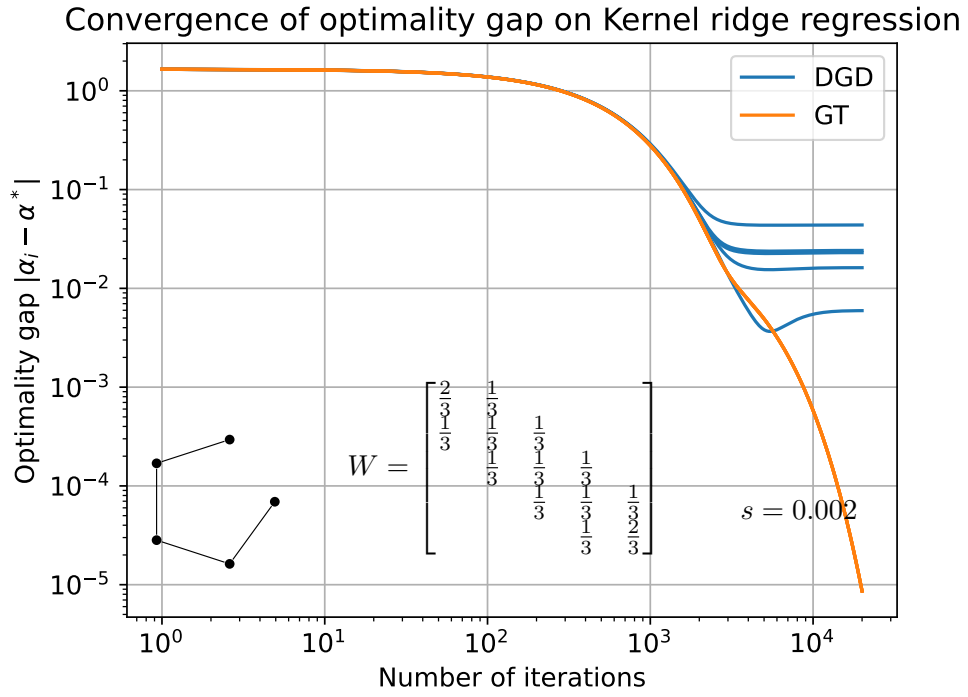


Figure 3: Convergence in terms of $\|\alpha_i - \alpha^*\|$ for DGD and Gradient tracking on the Kernel ridge example for a line graph.